format is ISO 8601, and if your data comes formatted that way as a string, it's easy to turn it into a date datatype in JavaScript, as you'll see in section 2.1.4. You'll work with dates and times often. Fortunately, both the built-in functions in JavaScript and a few helper functions in D3 are available to handle data that's tricky to measure and represent.

Although the cities dataset has no temporal data, keep in mind that temporal data for common entities like cities and countries is often available. In situations where you can easily expand your dataset like this, you need to ask yourself if it makes sense given the scope of your project. In contrast, the tweets data has a string that conforms to RFC 2822 (supported by JavaScript for representing dates along with ISO 8601) and can easily be turned into a date datatype in JavaScript.

RAW

Raw, free, or unstructured data is typically text and image content. Raw data can be transformed by measuring it or using sophisticated text and image analysis to derive attributes more suited to data visualization. In its unaltered form, raw data is used in the content fields of graphical elements, such as in labels or snippets.

The city names provide convenient labels for that dataset, but how would we label the individual tweets? One way is to use the entire content of the tweet as a label, as we'll do in chapter 5, but when dealing with raw data, the most difficult and important task is coming up with ways of summarizing and measuring it effectively.

2.1.3 Transforming data

As you deal with different forms of data, you'll change data from one type to another to better represent it. You can transform data in many ways. Here we'll look at casting, normalizing (or scaling), binning (or grouping), and nesting data.

CASTING: CHANGING DATATYPES

The act of casting data refers to turning one datatype into another from the perspective of your programming language, which in this case is JavaScript. When you load data, it will often be in a string format, even if it's a date, integer, floating-point number, or array. The date string in the tweets data, for instance, needs to be changed from a string into a date datatype if you want to work with the date methods available in JavaScript. You should familiarize yourself with the JavaScript functions that allow you to transform data. Here are a few:

```
parseInt("77");
parseFloat("3.14");
Date.parse("Sun, 22 Dec 2013 08:00:00 GMT");
text = "alpha,beta,gamma"; text.split(",");
```

Casts the string 77 into the number 77 with no decimal places

Casts the string 3.14 into the number 3.14 with decimal places

Casts an ISO 8601– or RFC 2822–compliant string into a date datatype

Splits the comma-delimited string into an array, which isn't strictly speaking a casting operation, but changes the type of data